Project FORESIGHT

Exploratory Data Analysis Insight Memo

Prepared for: NorthBay Living
Purpose: Summarise the strongest validated EDA findings and practical next steps.

Analysis Period	2011-01-29 to 2016-04-24
Integrated Records	573,900
SKU-Store Series	300
Inventory Basis	Simulated latest snapshot dated 2016-04-24

Executive Overview

EDA identified concentrated product demand, stronger weekend activity, meaningful category and geographic differences, mixed demand-driver relationships, and simulated inventory imbalance. These findings provide the analytical foundation for feature engineering, forecasting, and later inventory-risk analysis.

Key Findings and Recommendations

1. Product Concentration - Immediate Priority

Insight: Demand is concentrated in a limited part of the product portfolio.

Evidence: The leading product generated 22.56% of units; 14 of 30 products accounted for 81.95% of demand.

Recommended response: Protect availability and forecasting quality for high-contribution products.

Source: Sections 8–9
Scope note: Concentration is based on historical demand.

2. Simulated Inventory - Immediate Priority

Insight: Simulated inventory value is concentrated in overstock and weak-demand positions.

Evidence: Overstock held 66.68% of simulated inventory value; 40 weak-demand overstock series had 59.01 median coverage days.

Recommended response: Prioritise these series for forecast-based overstock assessment in Notebook 07.

Source: Section 11
Scope note: Inventory records and values are simulated.

3. Demand Behaviour - Near Term Priority

Insight: Weak and intermittent demand is meaningful, although no strict dead stock was found.

Evidence: 79 of 300 series were intermittent or very low/inactive, and 63 series entered the weak-demand watchlist.

Recommended response: Test specialised forecasting treatment and monitor weak-demand series separately.

Source: Sections 9 and 11
Scope note: Behaviour segments use relative portfolio thresholds.

4. Demand Timing - Near Term Priority

Insight: Demand is materially stronger on weekends.

Evidence: Weekend average demand was 24.43% higher than weekday demand.

Recommended response: Include weekly seasonality in forecasting and demand-planning workflows.

Source: Section 7
Scope note: Historical pattern requiring future validation.

5. Store and Geography - Near Term Priority

Insight: Performance differs materially across stores and states.

Evidence: California contributed 47.46% of demand and 48.32% of revenue; CA_3 led store performance.

Recommended response: Use store- and state-aware forecasting and inventory allocation.

Source: Section 8
Scope note: Results reflect the selected development scope.

6. Demand Drivers - Monitor Priority

Insight: Observed demand-driver relationships are mixed.

Evidence: Median price correlation was -0.02; event demand differed by -4.81%; SNAP demand differed by 4.94%.

Recommended response: Retain price, event, SNAP, and promotion features for time-based forecasting tests.

Source: Section 10
Scope note: Associations are not causal; promotion coverage was only 0.37%.

7. Category Performance - Strategic Priority

Insight: Demand-volume and revenue leadership differ.

Evidence: FOODS generated 41.0% of units, while HOUSEHOLD generated 41.7% of revenue.

Recommended response: Evaluate category decisions using both volume and revenue measures.

Source: Section 8
Scope note: Revenue contribution does not equal profitability.

Assumptions and Limitations

- The M5 retail data is adapted to the fictional NorthBay Living business case.

- The analysis uses a 300 SKU-store development scope.

- Inventory records are simulated; inventory value, cost, and margin measures are estimated or assumption-based.

- Promotion indicators are derived, and promotional records cover only 0.37% of observations.

- Price, event, promotion, and SNAP findings describe historical associations rather than causal effects.

- Final risk scores, reorder quantities, markdown actions, and financial exposure require forward forecasts in Notebook 07.

Next Analytical Steps

- Engineer leakage-safe calendar, price, promotion, event, SNAP, lag, and rolling features in Notebook 03.

- Evaluate stable, volatile, and intermittent series using appropriate forecasting methods.

- Compare forecasting models against the seasonal-naive baseline using time-based backtesting.

- Develop final stockout risk, overstock risk, reorder quantities, markdown actions, and financial exposure in Notebook 07.